2025 12th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO) : Submission (1080) has been created. Recibidos ×

Microsoft CMT <noreply@msr-cmt.org>
para mí

8:57 a.m. (hace 7 minutos)

Hello,

The following submission has been created.

Track Name: ICRITO2025

Paper ID: 1080

Paper Title: Comparative Analysis of Particle Swarm Optimization (PSO) and Optuna for Academic Performance Prediction in Peruvian Basic Education

Abstract:
Abstract—This article presents a comparative methodology for predicting student academic performance through Particle Swarm Optimization (PSO) applied to hyperparameter tuning of machine learning models. The study utilizes data from Peru's 2022 Ministry of Education evaluation, comprising 7839 sixth-grade student records with reading and mathematics scores plus socioeconomic variables from 25 regions. PSO is compared with Bayesian optimization using Optuna through 5-fold cross validation. Results demonstrate statistically equivalent performance. In reading, PSO achieved MSE of 4872 ± 106 and R^2 of 0.203 ± 0.016, while Optuna achieved MSE of 4872 ± 112 and R^2 of 0.203 ± 0.016. In mathematics, PSO obtained MSE of 6650 ± 261 and R^2 of 0.154 ± 0.021, compared to Optuna's MSE of 6651 ± 262 and R^2 of 0.154 ± 0.021. Student's t-tests confirmed no significant differences ($p > 0.05$). Efficiency analysis revealed PSO used 88 trees versus 177 for Optuna in reading, while Optuna used 152 trees versus 180 for PSO in mathematics. Territorial analysis showed coastal departments like Callao (575.29 points) significantly surpassing Amazonian regions like Loreto (453.50), creating a 121.79-point gap equivalent to 1.7 years of schooling. Both methods represent equivalent alternatives for educational hyperparameter tuning, with selection depending on computational resources.
Index Terms—Educational data mining, academic performance prediction, particle swarm optimization, hyperparameter tuning, machine learning.

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Secondary Subject Areas: Not Entered

Submission Files:
articulo_of.pdf (1 Mb, Thu, 24 Jul 2025 13:45:48 GMT)

Submission Questions Response: Not Entered

Thanks,
CMT team.